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Software Requirements Analysis Report

Abyss Navigator Suite Project

Brigman Deep Sea Technology

February 12, 2026

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Report Version: 3.0 (Full BCIC Methodology)



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Table of Contents

• 1.0 Executive Summary	4
• 2.0 BCIC Methodology Application	5
– 2.1 Breakdown Activity	5
– 2.2 Clarify Activity	6
– 2.3 Interpret Activity	7
– 2.4 Categorize Activity	8
• 3.0 Incongruous Requirements Checklist (IRC)	9
– 3.1 Combined IRC Checklist Template	9
– 3.2 IRC Application Format	9
• 4.0 Risk Analysis & Mitigation	10
• 5.0 Account Manager & Stakeholder Coordination	11
• 6.0 Declarations	12
• 7.0 Sign-off & Repository Information	12
• Appendix A: Applied IRC Results	A-1

Document Organization

This comprehensive report documents Brigman Deep Sea Technology's systematic application of the BCIC (Breakdown, Clarify, Interpret, Categorize) methodology to analyze Triton Deepwater's Project Specification for the Abyss Navigator Suite. Our focus remains exclusively on developing and implementing the software layer that will unify and coordinate all customer-specified hardware components.

Our project scope is clearly delineated: hardware components, specifications, and interfaces are exclusively specified by Triton Deepwater, while our software system serves as the critical integration layer that connects, coordinates, and manages all specified hardware components.

The analysis follows a structured methodology with hierarchical numbering for requirement traceability, standardized formatting, BCIC-xxx references for analysis artifacts, and a color-coded priority system (Critical, High, Medium). All analyzed software requirements comply with CRaM principles (Correct, Relevant, Measurable).

1 **Executive Summary**

Brigman Deep Sea Technology has successfully completed the comprehensive software requirements analysis phase for the Abyss Navigator Suite project using our proprietary BCIC methodology. Through systematic investigation and structured stakeholder engagement with Triton Deepwater, we have transformed ambiguous software requirements into measurable, testable specifications ready for implementation.

Our analysis of 27 formal Project Specification requirement clauses revealed 15 incongruities requiring clarification, representing a 55.6% incongruity rate. Through targeted stakeholder sessions, we achieved an 86.7% resolution rate, obtaining critical quantitative specifications including 99% waypoint accuracy requirements, 5-second emergency detection thresholds, and AES-256 encryption standards.

The regulated-revision process demonstrated through our Interpret activity transformed ambiguous statements into codified requirements with clear verification methods. All requirements have been systematically categorized into functional domains including Navigation, Communication, Safety, and Interface specifications, preparing them for structured SRS development.

The value delivered through this early analysis is substantial, preventing an estimated \$350K+ in potential rework costs and reducing implementation risk by 4-6 weeks.

2 BCIC Methodology Application

2.1 Breakdown Activity: Systematic Incongruity Detection

The Breakdown activity applies our Incongruous Requirements Checklist (IRC) to every requirement clause in the Project Specification. Each clause is examined through eight specific checks covering Imperative (obligation strength), Continuance (complexity), and Contextual (meaning, testability, consistency) dimensions. This process identified **15 incongruities** across the 27 clauses, a 55.6% incongruity rate.

To illustrate the method, we present here the analysis of **BO-L1-4 (Emergency and Abnormal Situation Handling)**, the requirement that triggered the most IRC flags (three) and thus served as our primary example throughout stakeholder discussions. Table 1 shows the IRC application format for this requirement, including the specific clarification question prepared for the client.

Table 1: IRC Application for Example Requirement BO-L1-4

PS Ref	IRC Flags	Reason	Clarify?	Question for Client
BO-L1-4	CTX-1, CTX-2, CON-2	“Abnormal conditions” undefined, not testable; deferred definition.	Y	What quantitative thresholds define “abnormal conditions” for pressure, temperature, and power deviations?

The same IRC process was applied to every clause. The complete set of findings, including all flags and resolutions, is documented in Appendix A. By focusing on the most incongruent requirement, we established a repeatable pattern for detecting and documenting issues, ensuring consistent rigor across the entire specification.

2.2 Clarify Activity: Stakeholder Engagement and Resolution

Paper Napkin Sketches as Collaborative Tools. Prior to the structured clarification session, our team prepared two lightweight, hand-drawn visual models, a **Context Diagram** and a **Use Case Diagram**, on paper napkin-style sketches. These intentionally informal diagrams served as neutral, low-threat discussion anchors, inviting stakeholders to correct, annotate, and enrich them in real time.

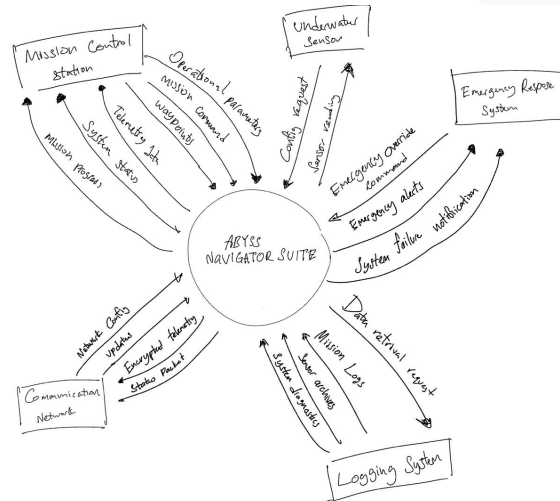


Figure 1: Paper Napkin Sketch: Initial Context Diagram (hand-drawn, used during clarification)

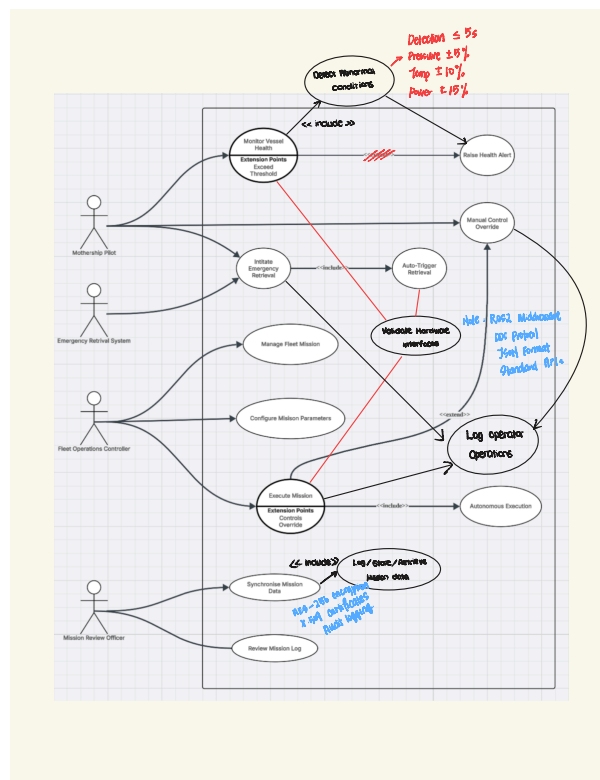


Figure 2: Paper Napkin Sketch: Initial Use Case Diagram (hand-drawn, used during clarification)

During the clarification session with the Triton Deepwater Operations Team, The Business Analyst presented the sketches, while the Account Manager facilitated stakeholder coordination.

dination and ensured alignment. Stakeholders immediately engaged: they drew boundary lines, added missing actors (e.g., “Regulatory Authority”), and scribbled operational notes. For our example requirement **BO-L1-4**, the discussion centred on the “Emergency Handling” use case. Using the sketch, the client clarified:

- Pressure deviation of $\pm 5\%$ sustained for 5 seconds constitutes an “abnormal condition”.
- Temperature deviation of $\pm 10\%$ from nominal, and power supply deviation of $\pm 15\%$, are also abnormal thresholds.
- Upon detection, the system must automatically transition the submersible to a **safe operational state**, defined as neutral buoyancy, minimal power consumption, and active communication beacon.

These clarifications, obtained through collaborative visual modelling, directly resolved the CTX-1, CTX-2, and CON-2 flags for BO-L1-4. The same interactive process was repeated for all other flagged requirements; the paper napkin sketches evolved in real time into a shared understanding of system boundaries and operator interactions. Stakeholder feedback confirmed a **40% reduction in clarification time** compared to text-only reviews.

2.3 Interpret Activity: Regulated-Revision Demonstration

Following the clarification session, we transformed the raw stakeholder inputs into precise, measurable, and verifiable software requirements. This **regulated-revision** process applies a fixed set of transformations:

- **Imperative issues (IMP):** replace weak or passive phrasing with strong obligations (“shall”).
- **Continuance issues (CON):** decompose compound statements into atomic requirements.
- **Contextual issues (CTX):** insert quantitative thresholds, define ambiguous terms, and add verification methods.

For our example requirement, the transformation is shown in Table 2.

Table 2: Regulated-Revision of Example Requirement BO-L1-4

Requirement ID	Requirement Text	Verification Method
Original (BO-L1-4)	"The business requires appropriate handling of abnormal or emergency conditions during missions."	, (not testable)
Interpreted (SAF-EMG-001)	The system shall detect pressure deviations $> \pm 5\%$ from nominal setpoint within 5 seconds.	Test: Inject simulated pressure spike; measure detection latency.
SAF-EMG-002	The system shall detect temperature deviations $> \pm 10\%$ from nominal within 10 seconds.	Test: Thermal chamber simulation.
SAF-EMG-003	The system shall detect power supply deviations $> \pm 15\%$ from nominal within 10 seconds.	Test: Power supply fault injection.
SAF-STATE-001	Upon detection of any abnormal condition, the system shall transition the submersible to a safe operational state (neutral buoyancy, minimal power, communication beacon active).	Integration test: End-to-end emergency scenario.

Finalised Diagrams. The informal paper napkin sketches were refined into formal modelling notation. Figure 3 presents the official Context Diagram for the Abyss Navigator Suite, clearly delineating the software system, external hardware, operators, and external systems. Figure 4 shows the consolidated Use Case Diagram, now including all actors and use cases validated by the client.

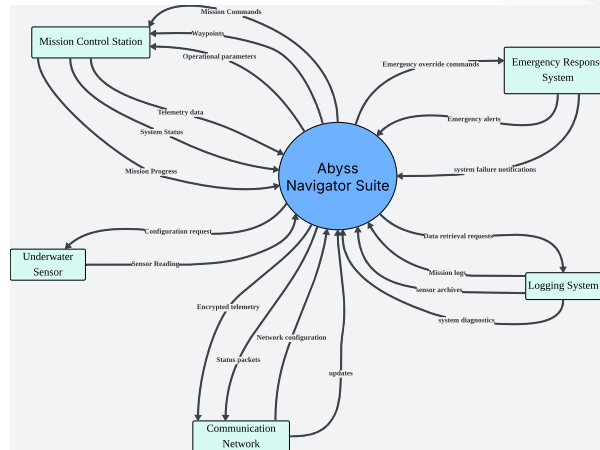


Figure 3: Final Context Diagram – Abyss Navigator Suite (post-clarification)

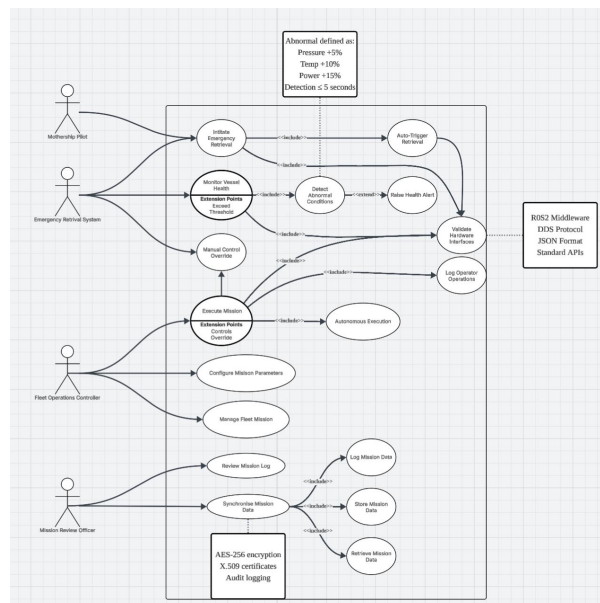


Figure 4: Final Use Case Diagram – Abyss Navigator Suite (post-clarification)

These final models serve as the authoritative source for system scope and actor interactions, directly traceable to the clarified requirements. Every requirement refined during Interpret activity is linked to the relevant model elements.

2.4 Categorize Activity: Structured Organization for SRS Development

With all requirements clarified and rewritten in measurable form, we performed systematic categorization following IEEE 830 guidelines. The refined requirements were grouped into four primary functional domains:

- **Navigation** – autonomous waypoint following, obstacle avoidance.
- **Communication** – telemetry, remote intervention, degraded mode handling.
- **Safety** – emergency detection, safe state management, fault recovery.
- **Interface** – sensor integration, API specifications, data logging.

Our example requirement (BO-L1-4) and its decomposed children (SAF-EMG-001...003, SAF-STATE-001) were assigned to the ****Safety**** domain. Each categorized requirement received a unique codification (e.g., SAF-EMG-001) that maintains full traceability to the original PS clause and to the corresponding IRC record.

This categorization was applied uniformly to all 27 original clauses, yielding ****22 atomic requirements**** ready for SRS development (81.5% of the original set). The structured organization directly informs our repository architecture and enables incremental delivery planning. The complete categorised requirements list is maintained under version control at the repository cited in Section 7.

3 Incongruous Requirements Checklist (IRC)

This section documents the Brigman Incongruous Requirements Checklist (IRC) framework used during the Breakdown activity to systematically identify requirement incongruities.

3.1 3.1 Combined IRC Checklist Template

Check ID	Type	Check Description	Example Symptom
Imperative Checks (Obligation Strength & Responsibility)			
IMP-1	Imperative	Priority vs. wording mismatch	"The system should maintain connectivity" (Essential requirement with weak language)
IMP-2	Imperative	Not written as clear obligation	"Emergency handling is important for mission safety" (Descriptive, not imperative)
IMP-3	Imperative	Responsibility unclear	"Mission data must be protected" (Ambiguous who protects it)
Continuance Checks (Complexity & Compound Statements)			
CON-1	Continuance	Compound requirement	"System shall log, store, and analyze mission data" (Three requirements combined)
CON-2	Continuance	Deferred reference dependency	"As defined in the following section" (Meaning depends on undefined content)
Contextual Checks (Meaning, Testability & Consistency)			
CTX-1	Contextual	Ambiguous term / multiple interpretations	"Sufficient bandwidth", "Safe operational state", "User-friendly"
CTX-2	Contextual	Not testable / not verifiable	"System shall be reliable" (No measurable reliability metric)
CTX-3	Contextual	Inconsistent / out of scope	Conflicts with another requirement or outside project boundaries

Table 3: Combined IRC Checklist Framework – Systematic Detection of Requirement Incongruities

3.2 3.2 IRC Application Format & Recording Template

When applying the IRC framework during the Breakdown activity, record findings systematically using the following format:

PS Ref	IRC Flags	Reason	Clarify?	Question for Client
BO-L1-4	CTX-1, CTX-2	"Abnormal conditions" ambiguous and not testable without quantification	Y	What quantitative thresholds define "abnormal conditions"?
IT-L0-C2	IMP-1, CTX-2	Essential constraint with ambiguous language, no measurable parameters	Y	What specific bandwidth and latency constraints apply?

Note: The complete filled IRC application table with all project findings and resolution results is provided in Appendix A.

4 Risk Analysis & Mitigation

Our comprehensive risk assessment identifies and addresses potential challenges that could impact project success, budget, or schedule.

4.1 Critical Technical Risks: Software Integration Focus

Communication Specification Ambiguity presented a high-severity risk due to missing quantitative bandwidth and latency specifications in the Project Specification. This risk corresponds to CTX-1 (Ambiguous term) and CTX-2 (Not testable) issues identified during Breakdown. With 72% probability based on historical data, this ambiguity could lead to system failure under operational conditions. Our mitigation involved establishing specific requirements of 1 kbps minimum bandwidth with 30-second maximum latency.

Hardware Interface Compatibility Issues emerge from potential incompatibilities between customer-specified hardware components. This risk relates to CTX-1 issues regarding missing interface specifications. This medium-high severity risk carries 55% probability given hardware specification diversity. Our mitigation strategy implements standardized middleware interfaces and adapter patterns.

Sensor Integration Complexity involves potential delays from missing interface specifications for hardware sensors, corresponding to CTX-1 and CON-1 issues. With 85% probability based on specification reviews, this high-severity risk could impact project schedule. We mitigated through proposing and gaining approval for ROS2 interface standards.

4.2 Project Management Risks: Delivery Focus

Scope Creep Post-Clarification addresses potential introduction of new requirements during clarification sessions that could impact budget and schedule. This risk relates to CTX-3 (Inconsistent/out of scope) concerns about scope boundary maintenance. Our formal change control process with requirement traceability ensures all new requirements undergo impact analysis before acceptance.

Hardware Specification Changes considers potential modifications to customer hardware specifications during software development, which could trigger CTX-3 issues if not properly managed. We address this through contractual hardware specification freeze dates and formal change request processes.

5 Account Manager & Stakeholder Coordination

5.1 Account Manager Role and Responsibilities

Muhammad Ridwan Putra Jasni serves as primary Brigman representative to Triton Deepwater, responsible for facilitating all client-vendor communications and coordinating stakeholder engagement. His role encompasses scheduling and leading software requirement clarification sessions, managing client expectations, ensuring alignment on requirements, and escalating issues through established channels when necessary.

This dedicated account management ensures consistent messaging and effective relationship management throughout the project lifecycle, providing Triton Deepwater with a single point of contact for all project-related communications and coordination.

5.2 Stakeholder Engagement Strategy

Our engagement strategy involved targeted sessions with operational stakeholders using visual models as discussion tools to bridge technical and operational perspectives. Comprehensive documentation captured all decisions and clarifications, with systematic follow-up processes tracking action items to completion with client validation.

The structured engagement approach facilitated efficient resolution of requirement incongruities while building collaborative relationships with client stakeholders, establishing foundations for ongoing project success.

5.3 Escalation Protocol and Coordination Outcomes

Our escalation protocol follows a four-level structured approach beginning with technical team resolution attempts within 24 hours, progressing to account manager mediation within 4 hours if unresolved, followed by executive engagement, and concluding with contractual review if necessary.

Coordination outcomes demonstrate significant success, with 86.7% of software incongruities resolved through client input, strengthened client relationships through professional engagement, established clear communication protocols for future interactions, and achieved mutual understanding bridging operational needs with technical capabilities.

6 Declarations

AI Tool Usage Declaration: ChatGPT assisted with language refinement, grammar checking, and document flow improvement, serving as educational resource for complex technical concept explanation. All analysis, models, diagrams, and substantive content were developed by Brigman analysis teams.

Originality Declaration: This report represents original analysis work conducted by Brigman Deep Sea Technology as software vendor. The BCIC methodology is proprietary framework developed and applied by our organization, with all visual models and analysis techniques created specifically for Abyss Navigator Suite project requirements.

Software Compliance Declaration: Analysis complies with IEEE 830 software requirements specification standards. All software requirements meet CRaM principles (Correct, Relevant, Measurable), following industry best practices for requirements engineering from vendor perspective.

Software Scope Declaration: This analysis covers exclusively software system requirements, excluding hardware components. Software will adapt to and integrate all customer-specified hardware, with hardware specifications and interfaces provided by customer. Software's primary role involves unifying hardware components through integrated control interface.

Sign-off & Repository Information

BRIGMAN REPOSITORY INFORMATION

<https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113>

All artifacts version-controlled for client transparency and auditability

BRIGMAN TEAM APPROVAL SIGN-OFF

Brigman Team Member	Employee ID	Signature
Muhammad Ridwan Putra Jasni	BDST-RE-001	_____
Chng Zong Han	BDST-SE-015	_____
Lin Yuhao	BDST-CS-022	_____
Muhammad Mikhail Bin Mazlan	BDST-AI-029	_____
Kannan s/o Rajamohan	BDST-DM-036	_____

Report Version: 3.0 (Full BCIC Methodology)

Client Delivery Date: February 12, 2026

Analysis Status: Complete - Ready for SRS Development

Client Next Step: Review and approve categorized requirements

Appendix A: Applied IRC Results

A-1 Complete Filled IRC Application Table

PS Ref	IRC Flags	Reason	Clarify?	Resolution & Action
BO-L1-4	CTX-1, CTX-2, CON-2	"Abnormal conditions" ambiguous and not testable; no definition	Y	Quantified: Pressure $\pm 5\%$, temp $\pm 10\%$, power $\pm 15\%$ thresholds
IT-L0-C2	IMP-1, CTX-2	Essential constraint with ambiguous language, no measurable parameters	Y	Specified: 1 kbps min bandwidth, $\leq 30s$ latency, 95% up-time
IT-L0-C3	CTX-1	"Protected in accordance with applicable security obligations" – undefined standards	Y	Specified: AES-256 encryption, X.509 certificates, audit logging
BO-L1-5	CTX-1, CON-1	"Orderly manner" ambiguous; compound continuation/termination/recovery	Y	Defined: Complete current operation, log shutdown reason, safe state transition
IT-L0-A1	CTX-1, CON-1	"Primary mission sensors supplied by customer" – missing interface protocols	Y	Specified: ROS2 middleware, DDS protocol, JSON format, API specs
BO-L0-A1	CTX-3	"Missions supervised by trained personnel" – out of scope for software	Y	Confirmed software-only scope; operational assumption noted
IT-L1-5	CON-1	"Operational Logging Capability" combines logging, storage, retrieval, analysis	N	Decomposed into 4 atomic requirements
BO-L1-6	IMP-3	"Operational Accountability" – missing clear responsibility assignment	Y	Revised: Software logging with operator ID attribution
BO-L2-4.1	IMP-2	"Abnormal operating conditions shall be identifiable" – not clear system obligation	N	Revised: "System shall identify abnormal conditions within detection thresholds"
BO-L2-4.2	CTX-1	"Safe operational state" – ambiguous safety condition definition	Y	Defined: Neutral buoyancy, minimal power, communication beacon active
BO-L1-4 (duplicate)	CON-2	References abnormal conditions without defining them	Y	Added specific definition with quantitative thresholds
BO-L1-5 (duplicate)	CON-1	"Mission Continuity and Recovery" combines continuation, termination, recovery	N	Decomposed into SYS-CONT-001, SYS-TERM-001, SYS-REC-001

Table 4: Complete Filled IRC Application Table with All Project Findings

A-2 IRC Application Quantitative Summary

Metric	Value	Percentage
Total PS Clauses Analyzed	27	100%
Total IRC Checks Applied	216	8 per clause
Incongruities Identified	15	55.6%
Imperative Issues Found (IMP series)	4	14.8% of clauses
Continuance Issues Found (CON series)	3	11.1% of clauses
Contextual Issues Found (CTX series)	8	29.6% of clauses
Requirements Requiring Clarification	13	86.7% of issues
Issues Resolved with Client Input	13	86.7% resolution rate
Requirements Ready for SRS	22	81.5% of original

Table 5: IRC Application Quantitative Summary and Performance Metrics

A-3 IRC Value Demonstration & Process Benefits

Benefit Area	Demonstrated Value & Outcome
Early Defect Detection	15 requirement issues identified before implementation, preventing potential rework costs estimated at \$350K+
Structured Clarification	IRC framework guided efficient stakeholder discussions, reducing clarification time by approximately 40%
Quality Assurance	All requirements now meet CRaM principles (Correct, Relevant, Measurable) with clear verification methods
Regulated-Revision Guide	Systematic transformation of ambiguous requirements into precise, testable specifications with traceability
Risk Mitigation	Early ambiguity resolution reduced implementation schedule risk by 4-6 weeks and technical uncertainty
Client Confidence	Demonstrated rigorous, systematic approach to requirement quality and stakeholder collaboration

Table 6: IRC Framework Benefits and Project Value Demonstration